MODEL LAB · 2026-10-28

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)	11	17,800	2.92%	0.5998
served prior	priors compared	held-out plate appearances	held-out HR rate	best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-13, scored on the 17,800 that came after (actual HR rate 2.92%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5813	1.60×	0.02826	0.13096	0.00454
KNN comps Prior is the pooled rate of the nearest comparables.	0.5998	1.64×	0.02821	0.12987	0.00605
SVR Support-vector regression on the same feature matrix.	0.5925	1.52×	0.02826	0.13049	0.00693
Random forest Forest regression; a learned similarity metric.	0.5951	1.50×	0.02825	0.13018	0.00678
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5998	1.64×	0.02821	0.12987	0.00605
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5998	1.64×	0.02821	0.12987	0.00605
Linear regression Ordinary least squares on the same features.	0.5964	1.58×	0.02825	0.13026	0.00487
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5953	1.58×	0.02830	0.13055	0.00609
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5955	1.58×	0.02824	0.13016	0.00503
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5945	1.45×	0.02829	0.13055	0.00733

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **260** knn **260** svr **260** rf **260** core **260** ensemble **260** linear **260** logistic **260** nn **260** xgb **260**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,620 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.6456	1.58×	0.02291	0.11126	0.01790
KNN comps Prior is the pooled rate of the nearest comparables.	0.6062	1.58×	0.02288	0.10980	0.01007
SVR Support-vector regression on the same feature matrix.	0.5783	1.58×	0.02323	0.11613	0.02309
Random forest Forest regression; a learned similarity metric.	0.6000	1.58×	0.02320	0.11319	0.02260
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6062	1.58×	0.02288	0.10980	0.01007
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6062	1.58×	0.02288	0.10980	0.01007
Linear regression Ordinary least squares on the same features.	0.6112	1.32×	0.02322	0.11383	0.02025
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6015	1.32×	0.02354	0.11442	0.02384
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6056	1.58×	0.02300	0.11128	0.01557
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5934	1.05×	0.02361	0.11640	0.02762

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jackson Chourio 15.3%, Junior Caminero 14.8%, William Contreras 14.0%, Jake Bauers 13.9%, Jonathan Aranda 11.6%, Christian Yelich 11.5%
Flat league	5/6	Jackson Chourio 15.5%, Junior Caminero 14.1%, William Contreras 13.9%, Jake Bauers 12.9%, Yandy Díaz 11.7%, Christian Yelich 11.5%
KNN comps	6/6	Jackson Chourio 15.8%, Junior Caminero 14.6%, William Contreras 14.1%, Jake Bauers 14.1%, Christian Yelich 11.8%, Jonathan Aranda 11.7%
SVR	5/6	Jackson Chourio 16.2%, Junior Caminero 15.9%, William Contreras 14.0%, Jake Bauers 13.8%, Yandy Díaz 11.9%, Jonathan Aranda 11.3%
Random forest	5/6	Jackson Chourio 16.1%, Junior Caminero 15.1%, Jake Bauers 14.1%, William Contreras 14.0%, Yandy Díaz 11.7%, Christian Yelich 11.5%
Core (KNN+SVR)	6/6	Jackson Chourio 15.8%, Junior Caminero 14.6%, William Contreras 14.1%, Jake Bauers 14.1%, Christian Yelich 11.8%, Jonathan Aranda 11.7%
Forest blend	6/6	Jackson Chourio 15.8%, Junior Caminero 14.6%, William Contreras 14.1%, Jake Bauers 14.1%, Christian Yelich 11.8%, Jonathan Aranda 11.7%
Linear regression	6/6	Jackson Chourio 16.0%, Junior Caminero 14.8%, William Contreras 14.0%, Jake Bauers 13.8%, Christian Yelich 11.7%, Jonathan Aranda 11.6%
Logistic regression	5/6	Jackson Chourio 16.0%, Junior Caminero 15.1%, William Contreras 14.0%, Jake Bauers 13.9%, Yandy Díaz 11.5%, Jonathan Aranda 11.4%
Neural net (torch)	5/6	Jackson Chourio 16.0%, Junior Caminero 15.1%, William Contreras 14.1%, Jake Bauers 14.0%, Yandy Díaz 11.5%, Jonathan Aranda 11.4%
XGBoost	5/6	Jackson Chourio 16.1%, Junior Caminero 15.4%, Jake Bauers 14.1%, William Contreras 13.9%, Yandy Díaz 11.8%, Christian Yelich 11.4%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1 Jackson Chourio MIL Slate	**15.3%**	15.5%	15.8%	16.2%	16.1%	15.8%	15.8%	16.0%	16.0%	
2 Junior Caminero TB Slate	**14.8%**	14.1%	14.6%	15.9%	15.1%	14.6%	14.6%	14.8%	15.1%	
3 William Contreras MIL	**14.0%**	13.9%	14.1%	14.0%	14.0%	14.1%	14.1%	14.0%	14.0%	
4 Jake Bauers MIL	**13.9%**	12.9%	14.1%	13.8%	14.1%	14.1%	14.1%	13.8%	13.9%	
5 Jonathan Aranda TB	**11.6%**	11.0%	11.7%	11.3%	11.3%	11.7%	11.7%	11.6%	11.4%	
6 Christian Yelich MIL	**11.5%**	11.5%	11.8%	11.2%	11.5%	11.8%	11.8%	11.7%	11.2%	
7 Yandy Díaz TB	**11.4%**	11.7%	11.5%	11.9%	11.7%	11.5%	11.5%	11.6%	11.5%	
8 Garrett Mitchell MIL	**10.6%**	10.2%	10.4%	9.9%	10.2%	10.4%	10.4%	10.2%	10.1%	
9 Brice Turang MIL	**10.3%**	10.4%	10.0%	10.2%	9.9%	10.0%	10.0%	9.6%	9.5%	
10 Ryan Vilade TB	**9.4%**	8.8%	9.3%	9.0%	9.3%	9.3%	9.3%	8.9%	8.8%	
11 Jonny DeLuca TB	**7.2%**	8.3%	7.3%	7.2%	6.9%	7.3%	7.3%	7.3%	7.6%	
12 Cooper Pratt MIL	**7.0%**	8.4%	6.9%	6.4%	6.2%	6.9%	6.9%	5.9%	6.1%	
13 Liam Hicks TB	**6.7%**	7.7%	6.7%	7.4%	7.3%	6.7%	6.7%	7.1%	7.0%	
14 David Hamilton MIL	**6.4%**	7.6%	6.3%	6.1%	5.7%	6.3%	6.3%	6.3%	6.2%	
15 Sal Frelick MIL	**5.5%**	7.0%	5.8%	5.2%	5.1%	5.8%	5.8%	5.6%	5.6%	
16 Ben Williamson TB	**4.4%**	7.5%	5.0%	4.4%	4.4%	5.0%	5.0%	3.1%	4.8%	
17 Taylor Walls TB	**3.6%**	4.8%	3.5%	2.6%	2.6%	3.5%	3.5%	3.3%	3.4%	
18 Chandler Simpson TB	**2.0%**	4.4%	2.6%	1.2%	1.3%	2.6%	2.6%	1.0%	2.2%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is a prior disagreeing about the hitter, not the context. Shading grows with the probability.

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